

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - FEB / MARCH 2025

B.E. - CSE (Cyber Security) /

Program	: CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Database Management Systems	Max. Marks	: 100
Course Code	: CY32 / CI32 (2021,2022,2023 Batch)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Draw the Entity- Relationship Diagram (ERD) for the following scenario: CO1 (07)
 - A salesperson may manage many other salespeople. A salesperson is managed by only one salespeople. A salesperson can be an agent for many customers.
 - A customer is managed by one salespeople. A customer can place many orders. An order can be placed by one customer. An order lists many inventory items. An inventory item may be listed on many orders.
 - An inventory item is assembled from many parts. A part may be assembled into many inventory items. Many employees assemble an inventory item from many parts. A supplier supplies many parts. A part may be supplied by many suppliers.
 - Differentiate Logical data independence and Physical Data Independence. CO1 (06)
 - Explain the three-schema architecture with a neat diagram. Justify the need of mappings between schema levels. Discuss how do different data definition languages support this architecture. CO1 (07)
- Draw ER Diagrams for the following statements with Structural constraints. CO1 (08)
 - A man can be married to at most one women and vice versa
 - An airport lies in exactly one country. A country may have arbitrarily many airports (and maybe none at all)
 - One instructor teaches many courses, but each course is run by exactly one instructor.
 - Each department has exactly one department head, some employees are the head of one department.
 - Describe the Degree of a relationship type and Recursive relationship with suitable examples. CO1 (06)
 - Discuss the characteristics of the DBMS Approach when comparing with the traditional file systems. CO1 (06)

UNIT - II

- Discuss the different types of update operations on a relation, indicating the possible violations of constraints. CO2 (06)

- b) Design the Complete Relational Schema for the Entity-Relationship diagram for MOVIE database in which data is recorded about the entire movie industry. CO2 (06)
- c) Given the schema : CO2 (08)
- EMP (Fname, Lname, SSN, Bdate, Address, Sex, Salary, SuperSSN, Dno)
- DEPT(Dname, Dnumber, MgrSSN, MGrstartdate)
- DEPT-LOC (Dnumber, Dloc) PROJECT(Pname, Pnumber, Ploc,Dnum)
- WORKS-ON (ESSN, PNo, Hours)
- Give the relation algebra expression for the following:
- (i) List female employees from Dno=20 earning more than 50000.
- (ii) List 'CSE' department details.
- (iii) Retrieve the name and address of all employees who work for sports department.
- (iv) Retrieve the name of the manager of each department.
4. a) Explain Join and Division Operations with suitable examples. CO2 (06)
- b) Using the following tables, evaluate the relational algebra operations listed below: CO2 (08)

Table A

Order Number	Company	City
A1002	Rentokil	London
A3333	Euros	Paris
B0987	Nokia	Glasgow
C7521	BT	Edinburgh
E0102	Halifax PLC	Halifax

Table B

Order Number	Company	City
E0102	Halifax PLC	Halifax
D2589	Videophone	London
B0987	Nokia	Glasgow

- (i) Union of A and B
- (ii) Difference A – B
- (iii) Difference B – A
- (iv) Intersection of A and B
- c) Consider the following relational schemas for a hotel management system: CO2 (06)
- Hotel(HotelID, HotelName, City, Rating)
 - Room(RoomID, HotelID, RoomType, Price)
 - Guest(GuestID, Name, City, Age)
 - Booking(BookingID, GuestID, RoomID, StartDate, EndDate)

Using relational algebra, solve the following queries:

- (i) List the names of all guests who stayed in hotels located in "Mysore" with a rating of 4 or higher.
- (ii) Find the names of guests and the total amount they spent on room bookings.
- (iii) Retrieve the names of hotels that have no bookings.

UNIT - III

5. a) Explain how SQL operations Union, Intersect and Except operate on relations with appropriate examples. CO3 (06)
- b) Differentiate between JOIN and NATURAL JOIN operations in SQL. CO3 (06)
- c) Person (driver-id, name, address) CO3 (08)
Car (license, make, model, year)
Accident (report-number, date, location)
Owns (driver-id, license)
Participated (report-number, license, driver-id, damage-amount)
Write SQL queries for the following
- Find the total number of people who owned cars that were involved in accidents in 2024.
 - Add a new accident to the database; assume any values for required attributes.
 - Delete the CAR belonging to "ANIL KUMAR".
 - List all the accidents that took place in location 'Mysore'.
6. a) Explain the purpose and syntax of the following commands in SQL by considering a relational schema that has at least one numerical attribute. CO3 (08)
i) Delete ii) Update iii) Alter iv) INSERT.
- b) STUDENT (regno: string, name: string, major: string, bdate: date) CO3 (12)
COURSE (course no: int, cname: string, dept.: string)
ENROLL (regno: string, course no: int, sem: int, marks: int)
BOOK _ ADOPTION (course no:int, sem: int, book-ISBN: int)
TEXT (book-ISBN: int, book-title: string, publisher: string, author: string)
Write SQL queries for the following:
- List the students name who have enrolled for 4th sem.
 - List no of books adopted in semester '2'.
 - List no of books adopted by 'AIML' department.
 - List the name of the student who has maximum enrollment
 - List the courses where course name begins with 'r' being held by 'cy' department.
 - List department which has more than 6 courses.

UNIT- IV

7. a) For a universal relation R(ABCDE), a set of FD's, F is given: {A→BC, CD→E, B→D, E→A, D→C}. Find the minimal cover of F, by illustrating the algorithm. CO4 (08)
- b) Prove or disprove the following inference rules for functional dependencies. CO4 (06)
Note: Read "I=" as implies
- {X→Y, Z→W} I= XZ→YW
 - {X→Y, XY→Z} I= X→Z
 - {XY→Z, Y→W} I= XW→Z
- c) Write an algorithm to find the closure of given functional dependency. CO4 (06)
Given the following set of functional dependency check whether they are equivalent or not. Relation R = (A, B, C, D, E, F)
- °F 1 = { A → C , AC → D , E → AD , E → F }
- °F 2 = { A → CD, E → AF }

8. a) Let Universal relation schema, $R = ABCDE$ with decomposed set of CO4 (07)
relations
 $R_1 = ABC$, $R_2 = BCD$, and $R_3 = CDE$.
Let the functional dependencies be: $AB \rightarrow CD$, $A \rightarrow E$, $C \rightarrow D$. Apply algorithm
to test whether the decomposition of R into $\{R_1, R_2, R_3\}$ is a lossless join
decomposition.
- b) Differentiate between 3NF and BCNF. Analyse whether the following CO4 (07)
relations are in 3NF or BCNF or both.
- Given relation $R(ABCD)$; with FD's $F = \{A \rightarrow B, A \rightarrow C, C \rightarrow D\}$
 - Given relation $R(XYZ)$; with FD's $F = \{XY \rightarrow W, W \rightarrow X, W \rightarrow YZ\}$
- c) Given the relation $r(R)$. State whether the following functional CO4 (06)
dependencies are satisfied by the relation or not.
- $X \rightarrow Y$
 - $XY \rightarrow Z$
 - $Z \rightarrow X$
 - $YZ \rightarrow X$

X	Y	Z
1	4	2
3	4	6
3	4	6
7	3	8
9	1	0

UNIT - V

9. a) Discuss about the Isolation levels of Transaction Processing. CO5 (06)
b) Explain the problems which may encounter when Transactions run CO5 (06)
Concurrently with examples.
c) Explain in detail about the ACID properties in Transaction Processing. CO5 (08)
10. a) Explain the distinction between the term's serial schedule and CO5 (06)
Serializable Schedule.
b) List and explain various failure classification in Recovery Systems. CO5 (06)
c) With a neat diagram write and explain the State Transition Diagram CO5 (08)
illustrating the states for transaction execution.
